A Jacobi-set based loss function for segmentation task

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Motivation

- Segmentation of fine-scale structures in natural and bio-medical images are gaining importance.
- The task still remains challenging as per-pixel accuracy is not only the metric of concern because of the imbalance in the dataset.
- Main motivation for this project comes from the paper titled "Topology-Aware Segmentation Using Discrete Morse Theory" by Hu et al. [HWF+21].
- A new loss function based on the Jacobi-sets [EH02] is proposed.

But what are Jacobi-sets?

Definition (Jacobi Sets)

Consider two Morse functions on a d dimensional manifold $\mathbb{M}$, $f, g : \mathbb{M} \to \mathbb{R}$. We denote the level set of g, $\mathbb{M}_t = g^{-1}(t)$.The *Jacobi-set* $\mathbb{J}(f, g)$ is the set of critical points of one function restricted to other's level sets.

$$\mathbb{J}(f, g) = cl\{x \in \mathbb{M} \mid \text{ x is a critical point of } f \text{restricted on } \mathbb{M}_t\}$$

- Intuitively *Jacobi-sets* can be thought as locus of critical points of f_t as g varies.

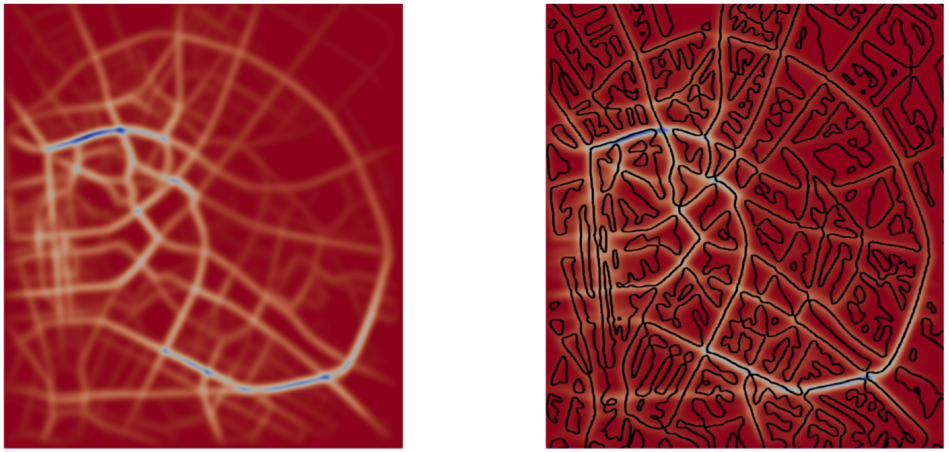

Figure: Jacobi sets

Computation for the PL Case

- x is a critical point iff $\nabla f + \lambda \nabla g = 0$ or $\lambda \nabla f + \nabla g = 0$
- Jacobi sets are closure of critical points of $f + \lambda g$ or $\lambda f + g$.
- Instead of keeping track of the critical points, in the PL setting we track edges along which critical points move.
- An edge uv is critical when
  - $h_\lambda(u) = h_\lambda(v)$ $[h_\lambda = f + \lambda g]$.
  - $h_\lambda(a) \cdot h_\lambda(b) > 0$ where $\{a, b\} = Lk(uv)$.

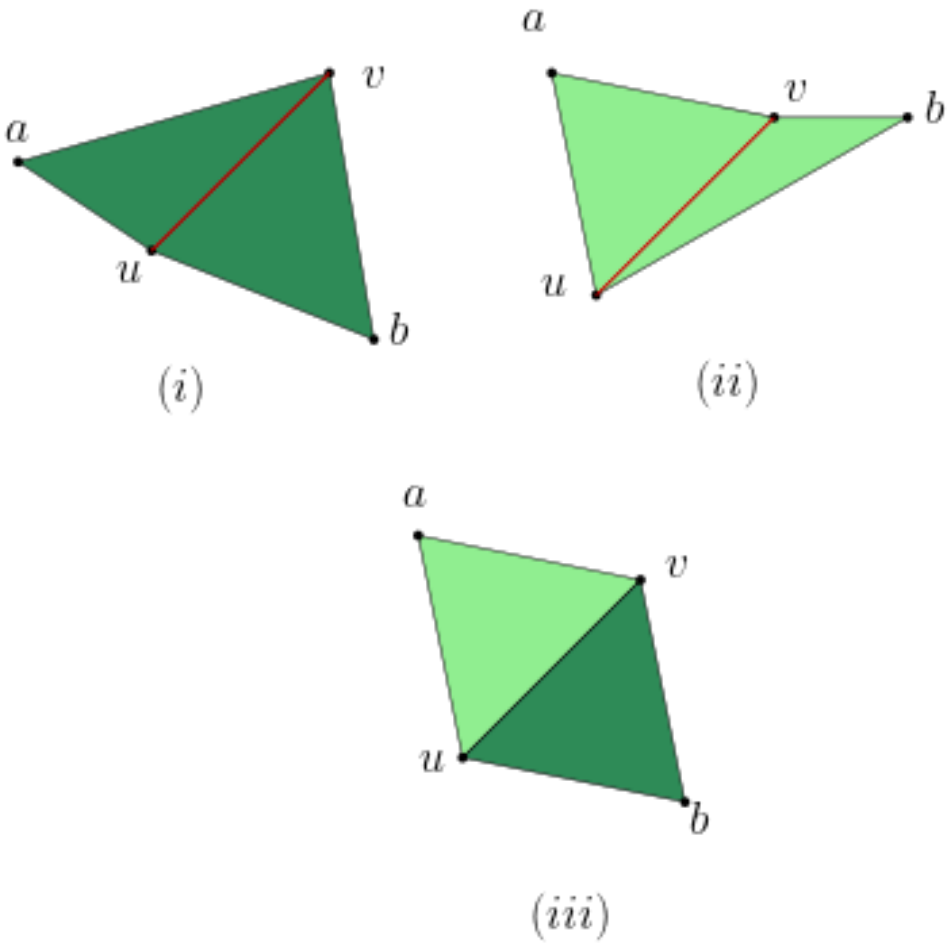

Figure: In (i) & (ii) edge uv is critical whereas in (iii), uv is regular.

Architecture

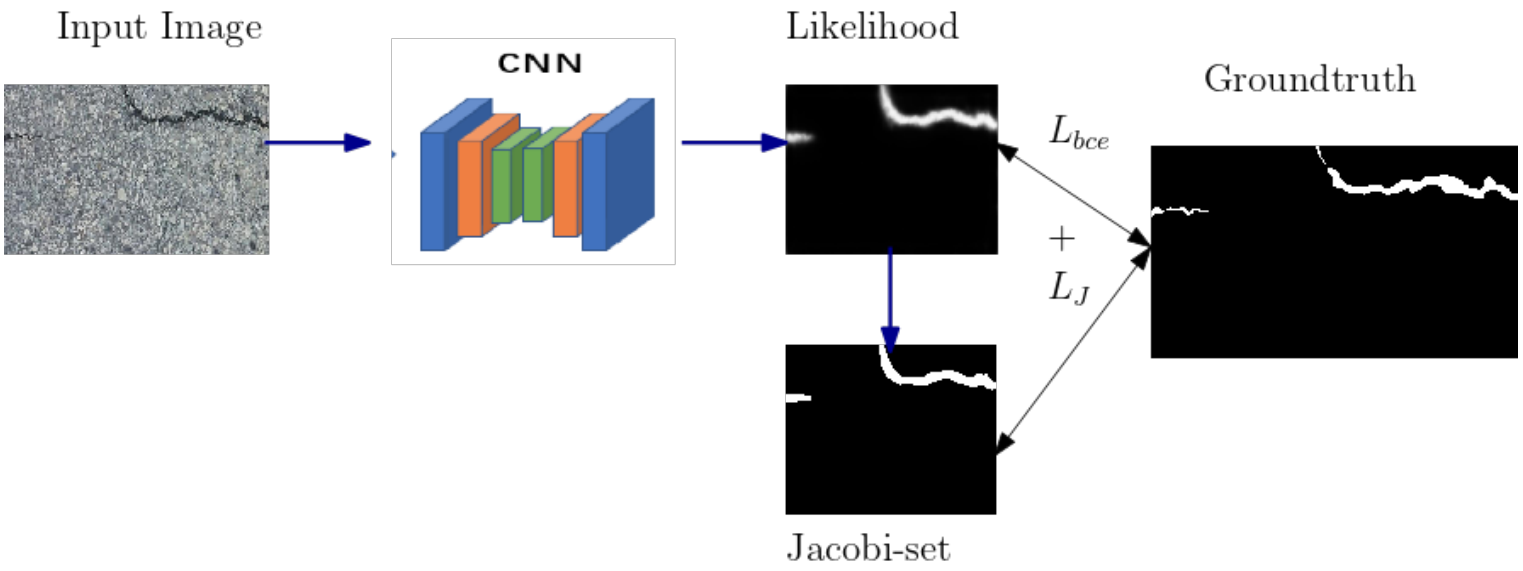

Figure: Overview of our loss function

Results

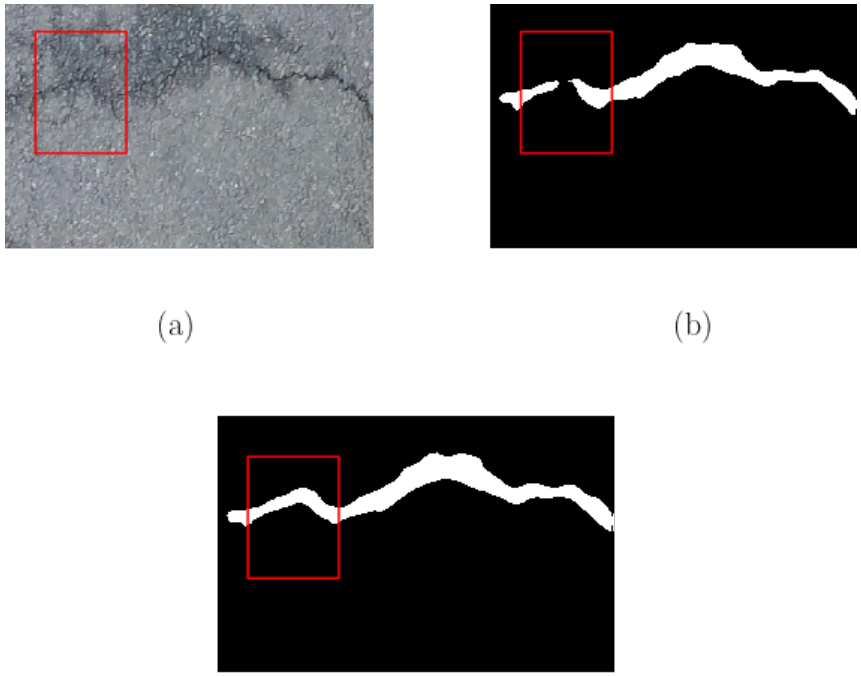

Figure: (a) Input image. (b) Segementation result with L_{bce} only. (c) Shows result with $L_{bce} + L_J$.

	Model with L_{bce}	Model with $L_{bce} + L_J$
DICE	0.657	0.616

Table: DICE score on CrackTree dataset with and without Jacobi loss.

Herbert Edelsbrunner and John Harer, *Jacobi sets of multiple morse functions*, Foundations of computational mathematics, Minneapolis **8** (2002), 35–57.

Xiaoling Hu, Yusu Wang, Li Fuxin, Dimitris Samaras, and Chao Chen, *Topology-aware segmentation using discrete morse theory*, arXiv preprint arXiv:2103.09992 (2021).